

WASP-10b

R-2444

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-10_120715 · created 2026-08-22T23:16:23+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2456123.8085 +/- 0.0015 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.204 +/- 0.062
Transit depth (Rp/Rs)^2	4.18 +/- 2.54 [%]
Orbital Inclination (inc)	86.16 +/- 1.86
Airmass coefficient 1 (a1)	0.84 +/- 0.46
Airmass coefficient 2 (a2)	0.09 +/- 0.17
Scatter in the residuals of the lightcurve fit is	53.53 %
Best Comparison Star	#1 - [551, 426]
Optimal Aperture	1.34
Optimal Annulus	5.25
Transit Duration (day)	0.066 +/- 0.032

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.204 ± 0.062 vs 0.1592 (deviation 0.72σ)
Duration fit vs published	0.066 d vs 0.0946 d (deviation 0.89σ)
Mid-time O–C	3.8 min
Depth SNR	3.3
Residual scatter	53.53 %

Light curve

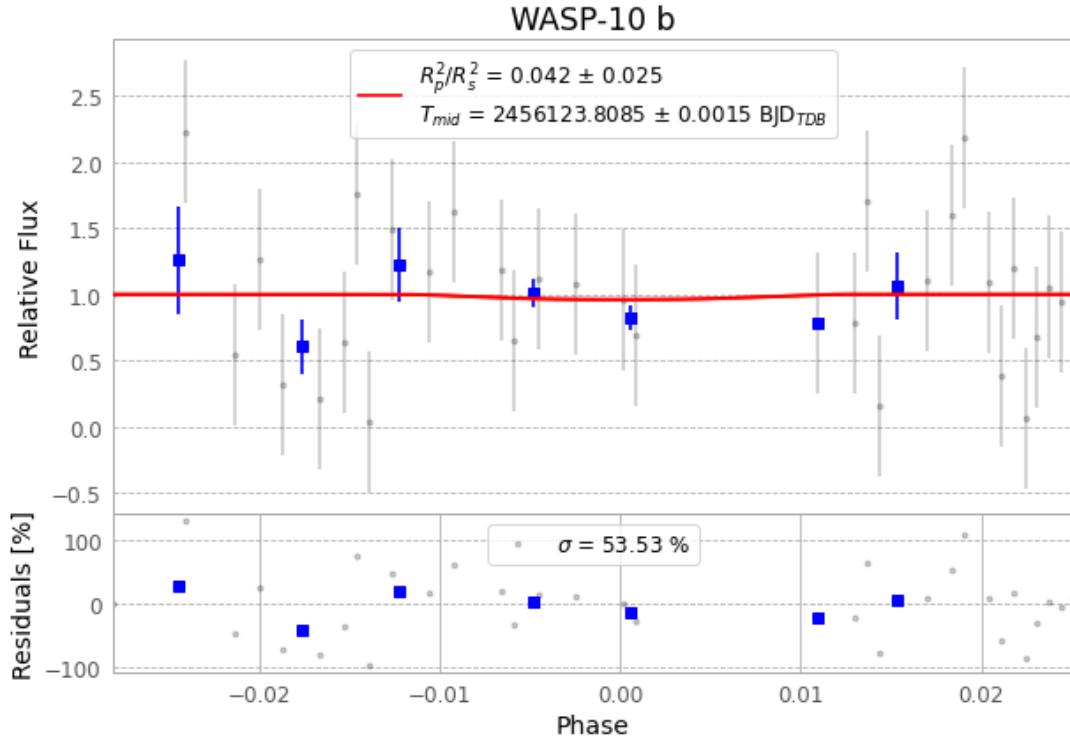


Figure 1 — FinalLightCurve_WASP-10 b_2012-07-14.png (SHA-256 b0a7d8c895158adb...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [265, 385], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 1333c954fade2ba7...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

81 FITS frames (53 MB), WASP-10120715051212.FITS → WASP-10120715091812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-10-120715.log (3d383aeab595...), FinalLightCurve_WASP-10 b_2012-07-14.png (b0a7d8c89515...), FinalLightCurve_WASP-10 b_2012-07-14.csv (ca78eb790bec...)

Compute resources

Wall time 133.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-22 23:16Z.